Advanced Numpy Project
Retail Sales & Business Analytics

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import numpy as np

import matplotlib.pyplot as plt


order_id=np.array([201,202,203,204,205,206,207,208,209,210,211,212,213,214,215])

category=np.array(["Electronics","Clothing","Grocery","Electronics","Home","Clothing","Grocery","Home","Electronics","Clothing","Grocery","Home","Electronics","Clothing","Home"])

unit_sold=np.array([12,25,40,8,15,30,55,10,5,18,35,20,7,22,12])

unit_price=np.array([15000,800,250,22000,3000,1200,180,4500,28000,1500,300,2500,18000,950,3800])

discount=np.array([10,5,0,15,10,5,0,12,8,10,5,8,5,0,10])

revenue=np.array([162000,19000,10000,149600,40500,34200,9900,39600,128800,24300,9975,46000,119700,20900,41040])

profit=np.array([24000,5000,1800,18000,7500,9000,1600,8200,16000,6200,1900,9500,14500,5500,8400])


print("=====================================Part A Sales Analysis=====================================")

print("======Units Sold Unit Price Adjusted For Discount======")

total=unit_sold*unit_price*discount

print(total)

print("======Order With Revenue Above 50000======")

print(order_id[revenue>=50000])

print("======Find Orders With Discount Greater Than 8 Percent======")

print(order_id[discount>=8])

print("======Calculate Average Revenue======")

avg=revenue.mean()

print(avg)

print("======Order Below Average Revenue======")

print(order_id[revenue<avg])
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print("=====================================Part B Productivity &
Profit=====================================")
print("======Calculate Profit Per Unit Sold======")
calcu=profit/unit_sold
print(calcu)
print("======Calculate Revenue Per Unit======")
cal=revenue/unit_sold
print(cal)
print("======Orders With Profit Above 8000======")
profit_abov=order_id[profit>=8000]
print(profit)
print("======Orders With More Than 20 Units Sold======")
orde=order_id[unit_sold>=20]
print(orde)
print("======Order With Profit Above 5000 and Discount Below 10%======")
find=(profit>=5000)&(discount<=10)
print(order_id[find])

print("=====================================Part C Performance
Analysis=====================================")
print("======Calculate Average Profit======")
avgp=profit.mean()
print(avgp)
print("======The Highest Profit======")
highest=profit.max()
print(highest)
print("======Lowest Revenue======")
lowest=revenue.min()
print(lowest)
print("======Median Revenue======")
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media=np.median(revenue)

print(media)

print("======Standard Deviation Of Revenue======")

st=np.std(revenue)

print(st)


print("=====================================Part D Order Classification=====================================")

print("======Order Revenue High to Low======")

reve=np.where(revenue>=10000,"High Value",np.where(revenue<=30000,"Medium Value","Low Value"))

print(reve)

print("======Classify Profit======")

p=np.where(profit>=10000,"Excellent",np.where(profit>=5000,"Good","Needs Review"))

print(p)

print("======Classify Discount======")

d=np.where(discount>=10,"High Discount","Regular Discount")

print(d)


print("=====================================Part E Management Insights=====================================")

print("======Identify High Revenue But Low Profit Orders======")

ident=(revenue>=50000)&(profit<20000)

print(order_id[ident])

print("======Identify High Discount Orders With Low Untis Sold======")

ident2=(discount>=10)&(unit_sold<=15)

print(order_id[ident2])

print("======Orders with Strong Profit======")

orde=(profit>=8000)&(unit_sold>=10)

print(order_id[orde])

print("======Identify Orders Profit Low & Discount HighNeeding Review======")

ident3=(profit<=3000)|(discount>=10)
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print(order_id[ident3])


print("=======================================Part F Mathematical Analysis=======================================")

print("======Round Profit Margin Percentage to 2 Decimal Places======")

de=np.round(profit,2)

print(de)

print("======Calculate Square Root Of Units Sold======")

sold=np.sqrt(unit_sold)

print(sold)

print("======Calculate Squared Revenue Values======")

sr=np.power(revenue,2)

print(sr)

print("======Calculate Absolute Difference Between Each Order's Revenue And Average Revenue======")

ab=np.abs(revenue)

print(ab)


print("=======================================Part G Final Business Report=======================================")

print("Net Revenue")

net=revenue-profit

print(net)

print("Profit Pr Unit Sold")

prof=profit/unit_sold

print(prof)

print("Profit Margin")

marg=(profit/revenue)*100

print(marg)

print("=====Count Revenue Values======")

count_h=np.sum(reve=="High Value")

print("Count High Value:",count_h)
```

```
count_m=np.sum(reve=="Medium Value")

print("Count Medium Values:",count_m)

count_l=np.sum(reve=="Low Value")

print("Count Low Value:",count_l)

print("=====Find Total Revenue Total Profit======")

print("Total Revenue:",revenue.sum())

print("Total Profit:",profit.sum())


print("=====================================Part H
Chart==================================")

plt.bar(category,profit,color='g')

plt.title("Category Wise Profit")

plt.xlabel("Category")

plt.ylabel("Profit")

plt.show()


plt.plot(category,profit,linestyle='-',marker='o',color='skyblue')

plt.title("Category Wise Profit")

plt.xlabel("Category")

plt.ylabel("Profit")

plt.show()


plt.figure(figsize=(7,7))

plt.pie(revenue,labels=category)

plt.show()


print("=====================================COMPETED=========================
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```